Domain-Specific Tokenization for Biomedical NLP

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Project Access

Repository:
notebooks + tokenizers + poster code

Study topic:
domain-specific tokenization

Motivation

- Biomedical text contains long compounds, abbreviations, and rare terms that are hard for generic tokenizers.
- Over-fragmentation increases sequence length, weakens entity boundaries, and may harm minority classes.
- Research question: does domain-specific tokenization improve both efficiency (fewer subtokens) and task robustness (classification + NER alignment)?

Main contributions

- We train two biomedical BPE tokenizers from PubMedQA text (`30k`, `50k requested / 39,681 actual`) with a reproducible setup.
- We evaluate intrinsic, downstream, and entity-aware tokenization behavior in one unified protocol.
- We show consistent gains over strong baselines (BERT-base, BioBERT, GPT-2) under a fixed classifier pipeline.

Hypothesis

Domain-trained BPE will reduce fragmentation in biomedical text and improve task-level robustness, especially for entity-heavy inputs.

Dataset Signals (PubMedQA)

Corpus Profile

Labeled: 1,000 **Unlabeled:** 61,249
Question length: mean 12.9 words
Context length: mean 200.2 words
Class balance: yes 55.2%, no 33.8%, maybe 11.0%

- **Source and task:** PubMedQA consists of biomedical question–context pairs with 3-way labels (*yes/no/maybe*) in the labeled split.
- **Data usage in this study:** unlabeled + labeled text for tokenizer training; labeled split for downstream classification.
- **Length asymmetry:** contexts are ∼15.5x longer than questions on average, so they dominate tokenization cost.
- Contexts dominate tokenization cost and drive model efficiency.
- Minority class (*maybe*) motivates macro-F1 over accuracy-only evaluation.

Experimental Setup

Baselines

BERT-base | BioBERT | GPT-2

Custom

BioBPE-30k | BioBPE-50k

Protocol: 300 samples for intrinsic analysis; PubMedQA 70/15/15 for classification; JNLPBA 4k/1k for NER metrics.

Tokenizer Adaptation Pipeline

1) Explore PubMedQA signals (length, rarity, class balance)
↓
2) Train biomedical BPE tokenizers (min_freq=10, 30k/50k)
↓
3) Intrinsic metrics: fertility, avg tokens, token length, UNK
↓
4) Downstream test: PubMedQA classification (fixed model)
↓
5) NER-oriented diagnostics on JNLPBA entities

Key design choice: only tokenization changes across configurations; model capacity and train/val/test split stay fixed for fair comparison.

Tokenizer Design in This Project

Design choice	Project setting
Algorithm	BPE from scratch (`tokenizers` library).
Normalization	NFKC + lowercase for robust lexical matching.
Pre-tokenization	Whitespace pre-tokenizer.
Special tokens	`[PAD]`, `[UNK]`, `[CLS]`, `[SEP]`, `[MASK]`.
Training text	PubMedQA context + question (labeled + unlabeled splits).
Vocabulary configs	30k and 50k; `min_frequency=10`.

Tokenizer Overview and Project Examples

Tokenizer families in NLP

Family	Typical behavior
WordPiece (BERT-style)	Greedy subword split with continuation markers; often higher fragmentation on rare biomedical terms.
BPE (our approach)	Merge frequent character/subword pairs; can retain domain morphemes when trained on in-domain corpora.
Byte-level BPE (GPT-2)	Strong open-vocabulary coverage, but biomedical lexical units may still split into many pieces.

Examples from this project

Term	BioBPE-50k	BERT-base
nephrolithiasis	[nephrolithiasis] (1)	[ne, ##ph, ##rol, ##ith, ##ias, ##is] (6)
gastroesophageal	[gastroesophageal] (1)	[gas, ##tro, ##es, ##op, ##ha, ##ge, ##al] (7)
electrocardiogram	[electrocardiogram] (1)	[electro, ##card, ##io, ##gram] (4)
immunohistochemistry	[immunohistochemistry] (1)	[im, ##mun, ##oh, ##isto, ##chemist, ##ry] (6)
hypercholesterolemia	[hypercholesterolemia] (1)	[hyper, ##cho, ##les, ##terol, ##emia] (5)
hepatocellular	[hepatocellular] (1)	[he, ##pa, ##to, ##cellular] (4)

Evaluation Signals

- **Segmentation compactness:** avg tokens per sample, fertility, token length.
- **Downstream utility:** test accuracy and macro-F1 on yes/no/maybe.
- **NER compatibility:** subtoken-per-entity ratio, single-piece entity rate, fragmentation rate.

Composite NER tokenization score (heuristic)

$$\text{Score} = 0.50(1 - \text{Frag}) + 0.30(\text{SinglePiece}) + 0.20(1 - \text{UNK})$$

Experimental Results

Stage 4: Intrinsic tokenization metrics (best to worst by fertility)

Tokenizer	Avg Tok/Sample	Fertility	Avg Tok Len
BioBPE-50k	273.26	**1.301**	5.12
BioBPE-30k	275.11	1.310	5.08
GPT-2	313.38	1.492	5.01
BERT-base	323.89	1.542	4.29
BioBERT	339.49	1.616	4.05

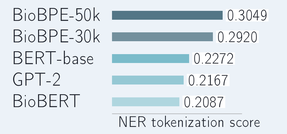
Relative gain: best custom model reduces fertility by ∼12.8% vs best baseline (GPT-2).

Stage 5: Downstream classification (test set)

Tokenizer	Accuracy	Macro-F1
BioBPE-50k	**0.5667**	**0.3718**
BioBPE-30k	**0.5667**	0.3710
GPT-2	**0.5667**	0.3643
BERT-base	0.5600	0.3582
BioBERT	0.5533	0.3521

Observation: custom BPE improves macro-F1 while keeping accuracy stable.

Stage 6: NER-oriented ranking



Key Takeaways

- Best tokenizer overall: **BioBPE-50k**.
- Intrinsic compactness and extrinsic quality improve consistently.
- Gains are strongest on entity-sensitive biomedical text (lower fragmentation, better single-piece entity rate).

Practical Deployment Guidance

- **Default:** BioBPE-50k (fertility 1.301, top macro-F1 0.3718, top NER score 0.3049).
- **If constrained:** BioBPE-30k with near-identical quality (macro-F1 gap 0.0008).
- **Gain vs best baseline (GPT-2):** ∼12.8% lower fertility and +0.0075 macro-F1.
- **Highest impact:** long biomedical contexts and multi-morpheme entities.

Adoption path: train in-domain BPE, validate on classification + NER diagnostics, then fine-tune full models.

Conclusions

Finding	Domain-adapted BPE consistently yields more compact biomedical segmentation.
Task effect	Custom tokenizers achieve top macro-F1 and best NER-oriented ranking.
Best config	50*k* vocabulary is the most stable across intrinsic + extrinsic evaluation.
Recommendation	Use custom biomedical BPE as default for PubMed-style pipelines.

Reliability Notes and Next Steps

Reproducibility now	6 staged notebooks, fixed seeds, and shared benchmark splits across all tokenizers.
Main threats	Single-seed variance, same-domain evaluation only (PubMedQA/JNLPBA), heuristic composite NER score.
Next experiments	Multi-seed confidence intervals, transformer fine-tuning, and cross-domain transfer tests.